Creating Convexity in a few steps.

Michel Talagrand

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Caratheodory's theorem and the main problem.

Theorem (Caratheodory)

Consider $A \subset \mathbb{R}^N$. *Then for each* $x \in \mathrm{conv} A$ *there exists* $B \subset A, \mathrm{card}\, B = N + 1$ *with* $x \in \mathrm{conv} B$.

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Can one create some convexity in a fixed number of steps, independent of the dimension?

The main problem

Reminder: γ_N standard Gaussian measure on $\mathbb{R}^N$.
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For $A \subset \mathbb{R}^N$ define $A + A = \{x + y; x, y \in A\}$.

Problem (PB 1)

Does there exist an integer n with the following property: Given any N and any balanced $A \subset \mathbb{R}^N$ with $\gamma_N(A) \geq 3/4$ the set $A + \cdots + A$ (n-times) contains a convex set C with $\gamma_N(C) \geq 1/2$.

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Difficulty for a positive answer: By which method are we going to construct C?

Difficulty for a negative answer: By which method are we going to construct a counterexample A?

Observation

Denote by B_N the Euclidean unit ball of $\mathbb{R}^N$.

Lemma

There exists $c > 0$ such that if $A \subset \mathbb{R}^N$ satisfies $\gamma_N(A) \geq 3/4$ then $cB_N \subset A - A$.

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Idea: if $\|x\| \leq c$ then $\gamma_N(A + x) > 1/4$ so that $A \cap (x + A) \neq \emptyset$ and $x \in A - A$.

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Recalling the Gaussian isoperimetric inequality, $A + \cdots + A$ (n times) has a measure very close to 1 for large n.

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Given $a > 0$ for large N the set

$$K_a = \left\{x \in \mathbb{R}^N; \forall t \in \mathbb{R}, |\frac{1}{N}\mathrm{card}\,\{i \leq N; x_i \leq t\} - \gamma_1((-\infty, t])| \leq a\right\}$$

has measure close to 1 for large N. Define
$A_a = \{\lambda x; |\lambda| \leq 1, x \in K_a\}$.

Theorem ($n = 2$ does not work)

Given L and $a, \varepsilon > 0$ for N large enough $L(A_a + A_a)$ does not contain a convex set C with $\gamma_N(C) > \varepsilon$.

Subgaussian r.v.s

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Definition

An $\mathbb{R}^N$ valued r.v. Y is subgaussian if for each $t \in B_N$ the real valued r.v. $Y_t = (t, Y)$ is subgaussian.

Gaussian and subgaussian

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Van Handel Comparison Theorem. There exists a universal constant L such that if Y is subgaussian and φ is convex then $\mathrm{E}\varphi(Y/L) \leq \mathrm{E}\varphi(G)$.

Problem (PB 2)

Does there exist an integer n such that for each $\mathbb{R}^N$-valued subgaussian r.v. X we can find standard Gaussian r.v.s $G_1, \dots, G_n$ such that $X = \sum_{i \le n} G_i$?

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More work of A. Song

Theorem (A. Song)

There exists a constant $c > 0$ such that if X is a centered real-valued subgaussian r.v. then cX is the sum of 3 standard gaussian r.v.s.

Define

$$C = \left\{ x \in \mathbb{R}^N ; |\sum_{i \leq N} x_i| \leq 10\sqrt{N}; \sum_{i \leq N} \exp x_i^2/4 \leq 3N/2 \right\}.$$

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This is a convex set, and $\gamma_N(C) \geq 3/4$.

Theorem (A. Song)

There is a number n such that whenever $A \subset \mathbb{R}^N$ is balanced and invariant by permutation of the coordinates and $\gamma_N(A) \geq 3/4$ then $C \subset L(A + \cdots + A)$ (n times.)

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We did not have to face the seemingly impossible problem to construct C!

The "Three Gaussian theorem" of D. Hua, A. Song, S. Tudose

Theorem (ThA)

If the $\mathbb{R}^N$ valued r.v. Y is such that $\mathrm{E}f(Y) \leq \mathrm{E}f(G)$ for each convex function f one can find three standard $\mathbb{R}^N$ valued r.v.s G_1, G_2, G_3 such that $Y = G_1 + G_2 + G_3$.

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Parenthesis: This is basically a complete description of all subgaussian r.v.s. By van Handel comparison theorem, if Y is

subgaussian Y/L satisfies the hypothesis of ThA. Conversely if $Y = G_1 + G_2 + G_3$ then $Y/3$ is subgaussian.

Theorem (ThB)

If $A \subset \mathbb{R}^N$ is compact symmetric and $\gamma_N(A) \geq 3/4$ then $5(A + A + A)$ contains a convex set C with $\gamma_N(C) \geq 3/4$.

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Dream: find a function $f(A, C)$ of a set A and a convex C and prove that if C maximizes this function then $C \subset 5(A + A + A)$.

Proof that ThC implies ThB

Theorem (ThC)

Consider a ball B centered at the orgin, and assume that $B \setminus (A + A + A) \neq \emptyset$ where A is compact symmetric with $\gamma_N(A) \geq 3/4$. Then there is a symmetric convex function f^* such that $f^*(0) = 0$, $f^* \geq 0$, $\mathrm{E} f^*(G) \leq 1$ and $x \in B \setminus (A + A + A) \Rightarrow f^*(x) \geq 1$.

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Proof that ThC $\Rightarrow$ ThB . Take a ball B large enough that $\gamma_N(5B) \geq 19/20$. Since $\mathrm{E} f^*(G) \leq 1$ the convex $C' = \{f^* < 5\}$

satisfies $\gamma_N(C') \geq 4/5$ so that if $C = 5B \cap C'$ we have $\gamma_N(C) \geq 3/4$.

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on $5(B \setminus (A+A+A))$, so that this set does not meet C'.

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An algorithmic construction of f^* is possible but unsatisfactory.

Proof of Theorem C: Reduction to the case A open.

Lemma

The set H of convex symmetric function f with $f \geq 0$, $f(0) = 0$ and $\mathrm{E}f(G) \leq 1$ is closed under the topology of uniform convergence on compact sets.

Reduction to the case A open: replace A by the set A_ε of points within distance $< \varepsilon$ of A, construct the functions f_ε^* and let $\varepsilon \to 0$.

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Let $K = B \setminus (A + A + A)$, a compact set.

Proof Th A implies Th C: Main Lemma

Lemma

Consider a probability measure μ on K, which is invariant under the map $x \mapsto -x$ and a r.v. Y of law μ. Then there is a convex function f such that $\mathrm{E}f(G) < \mathrm{E}f(Y)$

Proof Indeed otherwise $\mathrm{E}f(Y) \leq \mathrm{E}f(G)$ for each convex function f. The three Gaussian theorem show that $Y = G_1 + G_2 + G_3$.

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Proof Indeed otherwise $\mathrm{E}f(Y) \leq \mathrm{E}f(G)$ for each convex function f. The three Gaussian theorem show that $Y = G_1 + G_2 + G_3$. But since $\gamma_N(A) \geq 3/4$ we have

$\mathrm{P}(Y \in A + A + A) \geq 1/4$. Impossible because Y is valued in K.

Continuation of proof

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Proof Consider the convex function f produced by the previous lemma, $\mathrm{E} f(G) < \mathrm{E} f(Y)$. We can assume $f(0) = 0$. Consider the function f^* given by $2f^*(x) = f(x) + f(-x)$.

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Replace f^* by λf^*.

End of Proof that Th A implies Th C

Consider the following two convex sets:
H the convex set of functions $f^* \geq 0$ with $f^*(0) = 0$, $\mathrm{E} f^*(G) \leq 1$.
$\mathcal{C}$ the set of probability measures on K that are invariant under the map $x \mapsto -x$.
On $H \times \mathcal{C}$ consider the following bilinear form: $\varphi(f, \mu) = \int f \mathrm{d}\mu$.
Then

$$1 \leq \inf_{\mu} \sup_{f} \varphi(f, \mu) = \sup_{f} \inf_{\mu} \varphi(f, \mu).$$

Thus we can find $f \in H$ such that $\int f \mathrm{d}\mu \geq 1$ for each μ.

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Thus we can find $f \in H$ such that $\int f \mathrm{d}\mu \geq 1$ for each μ. If $x \in K$ take $\mu = 1/2(\delta_x + \delta_{-x})$ to see that $f(x) \geq 1$.

Elements of proof of Theorem A: The miraculous part.

Consider Y taking only finitely many values y_i, $i \leq n$. $\mathrm{P}(Y = y_i) = p_i > 0$. Assume that $\mathrm{E}\varphi(Y) \leq \mathrm{E}\varphi(G)$ for each convex function φ.

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$$\mathrm{E}q_i(G) = p_i \ ; \ \mathrm{E}Gq_i(G) = \frac{1}{2}x_i.$$

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Trick: we may assume $A_n = 0$. Look what happens when $\sum_{i\leq n} |b_i| + \|c_i\| = R$.

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Remember that $EW_i = y_i$

$$\mathrm{E} W_i - y_i = X_i + Z_i.$$

End of proof of Theorem A

Equality in law

$$G = \sum_{i \le n} W_i 1_{\{Y = y_i\}} = \sum_{i \le n} y_i 1_{\{Y = y_i\}} + \sum_{i \le n} (W_i - y_i) 1_{\{Y = y_i\}}.$$

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Each piece is Gaussian!

$$G = Y + G_1 + G_3$$

G_1, G_2 are Gaussian.

An abstract discrete setting

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We will use the space 2^N, the set of subsets of $\{1, \dots, N\}$ provided with the probability μ_p. A random element of 2^N is a set constructed by deciding if each $i \leq N$ belongs to the set at random, with probability p, and independently over i.

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A class A is *decreasing* if $I \in A, J \subset I \Rightarrow J \in A$ and is *increasing* if $I \in A, I \subset J \Rightarrow J \in A$.

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Definition

A subset A of 2^N is called p-small if $A \subset \bigcup_m H_{I_m}$ with $\sum_m \mu_p(H_m) = 1/2$.

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Problem (PB 3)

Does there exist an integer n such that whenever $\mu_p(A) \geq 3/4$ the set $A^{(n)}$ is p-small?

A partial result

ThB gives a partial result about PB3 (with significant extra work).

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The case $n = 2$ of PB3 seems to be crucial.

Problem

Does there exist $\alpha > 0$ such that whenever $\mu_p(A) \geq 3/4$ the set $A^{(2)}$ is (αp)-small?

A supporting example

Example to support a positive answer: Assume that pN is an integer. Let $A = \{I \in 2^N; \operatorname{card} I \leq Np\}$ so that $\mu_p(A) \simeq 1/2$.

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Let $m = Np + 1$. Recall that

$$\binom{N}{m} \leq \left(\frac{eN}{m}\right)^m.$$

Now

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is very small as soon as $n \geq 4$. But this example is somewhat misleading because the complement of A itself is already αp small.

A deep result in the right direction

Consider a sequence $t = (t_i)_{i \leq N}$, and define the function X_t on 2^N by $X_t(I) = \sum_{i \in I} t_i$. Think of this as a r.v.

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A probability measure ν on 2^N is *p-spread* if $\nu(H_I) \leq p^{\operatorname{card} I}$ whenever $I \in 2^N$.

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Recall that the support of ν is the collection of $I \in 2^N$ such that $\nu(\{I\}) > 0$.

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Recall that the support of ν is the collection of $I \in 2^N$ such that $\nu(\{I\}) > 0$.

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A weaker notion of smallness

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Definition

A class is weakly p-small if it does not carry a p-spread probability measure.

A set of small measure which is not weakly small.

Think of $\{1, \dots, N\}$ as made of k blocks of length a.

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However the uniform measure on A is $1/q$ spread. A is not $1/q$-weakly small, but $\mu_p(A)$ can be very small.

A daring conjecture

Problem

Does there exist $\alpha > 0$ *such that a weakly p-small class is* $(\alpha p) -$ *small ?*

After much very deep work, the best currently know result is as follows

Theorem (J. Park after main contribution of Hyu Tuan Pham)

If a class is weakly p-small it is weakly $\alpha p / \log\log(100/p)$ *small.*

Chen Li's breakthrough

Theorem (Chen Li)

If $\mu_p(A) \geq 1/2$, $A^{(2)}$ is weakly p-small.

Corollary

Consider a class $\mathcal{X}$ which supports a p-spread measure. Then for each set A with $\mu_p(A) > 1/2$ there is a set in $\mathcal{X}$ which is covered by two elements of A.

Nontrivial example. $N = M^2$, think if $\{1, \dots, N\}$ as $\{1, \dots, M\}^2$, consider the class of sets which meet each column and each line. It carries a L/M spread measure.

Indications on the proof.

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$$h_i(x,y) = 1 + \lambda \chi(x_i)\chi(y_i)$$

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Proof: There is some $i \in I$ with $x_i = y_i = 0$, so $h_i(x,y) = 0$.

Continuation of proof.

Thus if $I \in A^{(2)}$ we have

$$\int \int 1_A(x) 1_A(y) h_I(x,y) \mathrm{d}\mu_p(x) \mathrm{d}\mu_p(y) = 0. \qquad (1)$$

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$a := \mu_p(A) = \sum_{J \in 2^N} a_J^2$ and $a_\emptyset = a$. Note that

$\sum_{J \neq \emptyset} a_J^2 = a - a^2$. Straight computation of the left-hand side of (1) gives

$$\sum_{J \subset I} a_J^2 \lambda^{|J|} = 0, \qquad (2)$$

where $|J|$ denotes the cardinality of J

end of proof

Since for $J \neq \emptyset$ we have $\lambda^{|J|} \geq -(1-p)/p^{|J|}$ we obtain

$$0 \geq a^2 - (1-p) \sum_{J \neq \emptyset, J \subset I} a_J^2 p^{-|J|}.$$

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This holds for all I. Assume now that $A^{(2)}$ supports a p-spread measure ν. Integrating the above in I with respect to ν and since $\nu(\{J; I \subset J|\}) \leq p^{|J|}$ gives

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You think this is clever? Wait for the next result.

Further results

Given an hereditary class A and $I \in 2^N$ define $\theta_I(A)$ as the measure of the projection of A on $\{0,1\}^N$ for the **uniform** probability on that set.

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Jinyoung Park fed Chen's argument to an AI agent and prompted it to obtain the proof of a strong form one of my old conjectures.

Theorem

When ν is p-spread,

$$\mu_p(A) \leq \prod_{I \in 2^N} \theta_I(A)^{\nu(\{I\})}.$$

Using the convexity of log it follows that

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Theorem

If $\mu_p(A) > 1/2$ the set $\{I \in 2^N; \theta_I(A) \leq 1/2\}$ is not weakly small.

This is stronger that saying the $A^{(2)}$ is not weakly small, because $I \in A^{(2)} \Rightarrow \theta_I(A) \leq 1/2$.

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This is stronger that saying the $A^{(2)}$ is not weakly small, because $I \in A^{(2)} \Rightarrow \theta_I(A) \leq 1/2$. Fact

$\mu_{1/2}(A) > 1/2 \Rightarrow A^{(2)} = 2^N$.

Beginning of proof

A is hereditary. Make a Fourier expansion:

$$1_A = \sum_{I \in 2^N} a_I \chi_I.$$

Define F on $\mathbb{R}^N$ by $F(z) = \sum_{I \in 2^N} a_I \chi_I(z)$.

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Let us fix a set $J \in 2^N$. Consider independent $\xi_i \in \{0, 1\}$ with $P(\xi = 1) = 1/2$. Consider β to be determined later, and define the r.v.s $Z_{i,J}$ for $i \leq N$ as follows:

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Fix β such that $EZ_{i,J} = p$.

Define $Z_J = (Z_{i,J})_{i \leq N}$

Lemma

$EF(Z_J) = a_\emptyset = a := \mu_p(A)$.

This is because $E\chi(Z_{i,J}) = 0$ for each i.

Cauchy-Schwarz!!

Let $Y_J = \{i \in J; \xi_i = 1\} = \{i \in J; Z_{i,J} = 1\}$.

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Use Cauchy-Schwartz to obtain

$$(\mathrm{E}F(Z_J))^2 \leq \mathrm{P}(Y_J \in A)\mathrm{E}F(Z_J)^2. \tag{3}$$

Moreover $\mathrm{P}(Y_J \in A) = \theta_J(A)$.

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$$\mathrm{E}F(Z_J)^2 = \sum_{I \subset J} a_I^2 \rho^{|I|},$$

Thus (3) means

$$a^2 \leq \theta_J(A) \sum_{I \subset J} a_I^2 \rho^{|I|}.$$

End of proof

Taking log:

$$2\log a \le \log \theta_J(A) + \log \sum_{I \subset J} a_I^2 \rho^{|I|}. \tag{4}$$

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If ν is p-spread and since $\rho p \le 1$

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Integrating (4) with respect to ν and using the convexity of the log gives

$$2\log a \le \int \log \theta_J(A) \mathrm{d}\nu(J) + \log a,$$

which is the desired inequality.